white fabric or coloured fabric. The dye is imprinted on the fabric in paste form and any desired pattern may be produced. Dark colour prints in lighter background is the characteristic feature of direct style printing. In this style of printing, the printing paste is transferred to the selected areas of the fabric and the pigments adhere to the fabric surface. Direct style of printing is used in block printing, screen printing or roller printing methods.

- easiest style of printing
- economical
- This style is easy, economical and suitable for printing both simple and complicated designs.

3.4.2 Discharge Style of Printing

In this method, the fabric is printed on a dyed fabric. The printing paste used in this method contains a discharging agent, which will bleach or destroy the colour from the dyed fabric in the printed areas. The resulting white area is brighten the overall design. Sometimes the base colour is removed and another colour is printed in its place. The discharging agent is an oxidizing or reducing agent capable of discharging colours by oxidation and reduction. Potassium chlorate or sodium chlorate (oxidizing agents) and stannous chloride (reducing agent) are commonly used discharging agents. The effects produced are very striking as the white area obtained brightens the overall design. This style of printing enabled intricate and fine designs to be printed on the fabric. If the fabric is not thoroughly washed after printing, the strength of the fabric may be affected due to the use of discharging agents. The advantages of discharge style of printing produces light, bright colour on a dark background, printing is sharp and fine and easier to work. However the major disadvantages of this method is the cost involved.

3.4.3 The Resist Style

In this method, the bleached fabric is first printed with a substance like wax, rice paste, china clay or chemicals such as acids, alkalis and salts that resist dye penetration and fixation. The printed fabric is then dipped in cold dye bath, so that the resisting agent remains unaffected and only the areas free of the resist agent are coloured. After dyeing process, the resist paste is removed, leaving white or light coloured patterns on a dark background. Batik, tie and dye are examples of resist printing. The durability of the fabric is not affected by the resist method.



Ancient Printing of Fabric

The earliest type of printing on fabric is block printing also called relief printing. This is the process of dye being pressed onto a fabric from a carved material; historically wood, copper but also rubber and now many other materials. Then in the 18th Century the technique of roller or cylinder printing came about, this is the process by which the fabric is carried along a rotating central cylinder and pressed by a series of rollers, each of which is engraved with the design. Each roller is fed a different colour through feed rollers, and some roller printing machines were even able to print 6 colours at once, making them much faster than the block printing process.



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